

Discovery Process and Proof of the $d = -427$ Ultra-Fast π Equation

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Abstract

In the field of computational mathematics, searching for ultra-fast converging Pi (π) formulas presents significant algebraic challenges. Traditionally, convergence limits for pure rational bases (class number $h = 1$) stopped at the Heegner number $d = -163$ (Chudnovsky formula, approximately 14.18 digits/term). This paper details how the "Quantity Free-Radical Cancellation" theory and dimensionality-reduced PSLQ algorithm were used to cross the class number barrier and discover an ultra-fast hypergeometric equation based on the discriminant $d = -427$ (class number $h = 2$). This equation achieves a convergence rate of 24.96 digits per term. This paper reproduces the entire discovery process and verifies the absolute correctness and engineering usability of the results through a lossless 100-million-digit stress test in a pure integer algebraic ring.

Keywords: Constant Computation, Hypergeometric Series, PSLQ Algorithm, Quantity Free-Radical Cancellation, Binary Splitting

1 Theoretical Background and Core Hypothesis

Since Srinivasa Ramanujan established the foundation for calculating π using hypergeometric series, improvements in computational speed have relied heavily on the algebraic properties of singular moduli. The Chudnovsky brothers, in 1988, utilized $d = -163$ (the largest imaginary quadratic field with class number 1) to construct the formula that has dominated the field for decades.

To break the 14.18 digits/term limit, mathematics must venture into imaginary quadratic field extensions with class number $h \geq 2$. We selected the discriminant $d = -427$ (i.e., 7×61), whose theoretical convergence rate represents a generational leap.

However, once entering higher-order class fields, traditional analytical methods fall into an algebraic quagmire. For $h = 1$, the Chudnovskys could derive pure rational numerator constants directly by differentiating the Klein j -invariant via complex analysis. For $h = 2$, the relevant parameters fracture into complex irrational numbers (residing in the ring of algebraic integers $\mathbb{Z}[\sqrt{61}]$). Forward derivation of modular form derivatives is not only extremely tedious but often results in massive nested radicals difficult to simplify into elegant closed forms. Simultaneously, because the numerator projections are no longer pure rationals, attempting to use traditional integer relation algorithms for a blind search among complex irrational coefficients encounters a severe curse of dimensionality.

Core Methodological Hypothesis: Quantity Free-Radical Cancellation

To solve this catastrophe, this paper proposes and applies the theory of "Quantity Free-Radical Cancellation." The essence of this theory is to treat the infinite series as a signal stream in a high-dimensional algebraic space. Since the irrational base resides in an algebraic extension field (such as $\mathbb{Q}(\sqrt{61})$), each term of the series generates residuals where rational and irrational parts are intertwined; we refer to these residuals as "Quantity Free-Radicals."

As long as we construct a precise multi-dimensional probe matrix in a homologous algebraic extension field, we can force these irrational residuals to undergo perfect cancellation (annihilation) at the physical precision level through extreme-precision lattice basis reduction algorithms (such as PSLQ), thereby directly extracting the pure algebraic integer weights hidden in the chaos.

2 Discovery Process and Methodology

2.1 Theoretical Foundation: Selection of Probe Engine and Base

Foundation 1: Combinatorial Series Engine $E(6n)$

We selected the Chudnovsky-class $E(6n)$ hypergeometric engine:

$$\text{Engine}(n) = \frac{(6n)!}{(3n)!(n!)^3} \quad (1)$$

Foundation 2: Irrational Base $C(427)$ and Introduction of $\mathbb{Q}(\sqrt{61})$

The convergence speed of the series is proportional to $\log_{10}(|C'|)$. In pure rational fields with $h = 1$, $d = -163$ constitutes the theoretical convergence limit. To break this limit, one must enter algebraic extension fields with $h \geq 2$. We identified $d = -427$, the largest discriminant in the imaginary quadratic field with $h = 2$.

According to Complex Multiplication (CM) theory, the modular j -invariant corresponding to $d = -427$ satisfies a specific quadratic equation. Modern computer algebra systems show that the actual modular j -invariant satisfies:

$$x^2 + 9034407125303115694336000x + 51923170459928424448000000 = 0 \quad (2)$$

The base C_{-427} required for calculating π also satisfies a specific quadratic equation:

$$x^2 + 15611455512523783919812608000x + 155041756222618916546936832000000 = 0 \quad (3)$$

Yielding the precise irrational base residing in $\mathbb{Q}(\sqrt{61})$:

$$C_{-427} = -(7805727756261891959906304000 + 999421027517377348595712000\sqrt{61}) \quad (4)$$

The absolute value of this base is 2.14×10^{25} , allowing a convergence rate of 24.96 digits per term.

2.2 Ultra-High Dimensional PSLQ Blind Search

We set the target as $\frac{\sqrt{|C_{-427}|}}{\pi}$. We used the hypergeometric engine to construct two independent signal streams S_0 and S_1 . To perform lattice basis reduction over $\mathbb{Z}$, we split the complex probes into rational and irrational parts, constructing a 5D probe vector $\vec{V}$:

$$\vec{V} = \left[\frac{\sqrt{|C_{-427}|}}{\pi}, \quad \text{Re}(S_0), \quad \text{Im}\left(\frac{S_0}{\sqrt{61}}\right), \quad \text{Re}(S_1), \quad \text{Im}\left(\frac{S_1}{\sqrt{61}}\right) \right] \quad (5)$$

PSLQ successfully locked onto a unique integer weight vector:

$$[-1, \quad 19885743328380, \quad 2546108530944, \quad 1290938757633000, \quad 165287770712064] \quad (6)$$

Extracting the common divisor 12, we parsed the algebraic forms of A and B :

- $A = 1657145277365 + 212175710912\sqrt{61}$
- $B = 107578229802750 + 13773980892672\sqrt{61}$

2.3 Reconstructed Ultra-Fast Equation

The final equation is:

$$\frac{1}{\pi} = \frac{12}{\sqrt{|C_{-427}|}} \sum_{n=0}^{\infty} \frac{(-1)^n (6n)!}{(3n)! (n!)^3} \cdot \frac{A + B \cdot n}{C_{-427}^n} \quad (7)$$

3 Proof of Methodology Rigor and Reliability

3.1 Mathematical Rigor

The core criterion for establishing an empirical formula as a strict algebraic identity lies in the Algebraic Isolation defined by the Liouville-Roth theorem. For the target and series vector $\vec{V}$, its inner product with any integer weight vector $\vec{W} \in \mathbb{Z}^5$, if not identically zero, must satisfy:

$$|\vec{V} \cdot \vec{W}| > 10^{-H} \quad (8)$$

Evaluation shows $H \ll 1000$. Our PSLQ reduction compressed the residual to below 10^{-1500} at 2000-digit precision. Since $10^{-1500} \ll 10^{-H}$, the residual must be mathematically zero.

3.2 Structural Rigor

The equation was implemented using a Binary Splitting architecture, which completely isolates cumulative truncation errors by merging all intermediate nodes as large integers and performing only one final high-precision division.

3.3 Engineering Verification

Stress tests using pure C with GMP achieved Bit-Perfect matching at precisions of 1M, 10M, and 100M digits, fully verifying the reliability of the formula.

4 Conclusion

This research confirms that π can be viewed as a linear projection of high-dimensional modular forms. Using the "Quantity Free-Radical Cancellation" system, we extracted the series equation for $h = 2$. We not only reproduced classic formulas such as Ramanujan (1914), Borwein (1987), and Guillera (2002), but also discovered new series such as the $\mathbb{Q}(\sqrt{6})$ series:

$$\frac{2}{\pi} = \sum_{n=0}^{\infty} \frac{((2n)!/(n!)^2)^3}{(-64(5 + 2\sqrt{6})^4)^n} \left[3(59 - 24\sqrt{6}) + 84(5 - 2\sqrt{6})n \right] \quad (9)$$

and the $\mathbb{Q}(\sqrt{3})$ $E(4n)$ series:

$$\frac{24}{\pi} = \sum_{n=0}^{\infty} \frac{(-1)^n \frac{(4n)!}{(n!)^4}}{(9216(2 + \sqrt{3})^4)^n} \left[(20\sqrt{3} - 27) + n(112\sqrt{3} - 84) \right] \quad (10)$$

among others not yet publicly published. This methodological system possesses high universality for the automated exploration of mathematical constants.